

GLOBAL SUPERSTORE ETL REPORT

Extract, Transform and Load Process

1. Introduction

This report documents the ETL (Extract, Transform and Load) process performed on the Global Superstore dataset. The dataset was processed using Python and Pandas in Google Colab. The main objective was to extract the raw data, profile its quality, perform appropriate transformations, load the cleaned dataset, and validate the final result.

2. Dataset Information

Property	Value
Dataset	Global Superstore
Source	Kaggle
Original Rows	51290
Original Columns	24
Final Rows	51290
Final Columns	30

3. Extract

The raw Global Superstore CSV file was uploaded to Google Colab and extracted into a Pandas DataFrame. The dataset required Latin-1 encoding during extraction because the original CSV contained characters that could not be decoded using UTF-8.

Item	Result
Input File	Global_Superstore2.csv
Encoding Used	Latin-1
Extraction Status	Successful
Rows Extracted	51290
Columns Extracted	24

4. Data Profiling

Data profiling was performed before transformation to understand the structure and quality of the raw dataset. The profiling process examined the dataset shape, column names, data types, missing values, duplicate records and statistical information.

Profiling Check	Result
Rows	51290

Columns	24
Duplicate Rows	0
Columns With Missing Data	1

Missing Value Details

Column	Missing Records
Postal Code	41296

5. Data Transformation

The following transformations were applied to improve the quality and analytical usefulness of the dataset:

- Duplicate records were identified and removed.
- Leading and trailing spaces were removed from text fields.
- Order Date and Ship Date were converted to datetime format.
- Date parsing was standardized using day-first format.
- Missing Postal Code values were replaced with 'Unknown'.
- Numeric columns were checked for invalid negative and zero values.
- Order Year was created from Order Date.
- Order Month was created from Order Date.
- Order Month Name was created from Order Date.
- Order Quarter was created from Order Date.
- Order Year-Month was created for monthly analysis.
- Profit Margin was calculated using Profit divided by Sales.

6. Business Data Validation

Numeric values were investigated to distinguish genuine business values from invalid data. Negative profit values were retained because they represent transactions where a loss occurred. Similarly, zero discount values were retained because they represent transactions where no discount was provided.

Check	Result
Negative Profit Records	12544
Zero Discount Records	29009

7. Load

After transformation, the cleaned DataFrame was exported as a new CSV file. The original dataset was preserved and the transformed data was saved separately.

Property	Value
Output File	Global_Superstore_Cleaned.csv

Rows Loaded	51290
Columns Loaded	30
Load Status	SUCCESS

8. Final ETL Validation

Validation	Result
Final Rows	51290
Final Columns	30
Duplicate Records	0
Remaining Missing Values	0
ETL Status	COMPLETED SUCCESSFULLY

9. Conclusion

The Global Superstore dataset was successfully processed through an end-to-end ETL pipeline using Python and Pandas in Google Colab. The raw data was extracted and profiled, data quality issues were identified and appropriate transformations were applied. The resulting cleaned dataset was then loaded into a new CSV file and validated to ensure the ETL process completed successfully.